REFINED VALUATIVE CRITERION FOR PROPERNESS WITH DENSE IMAGE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2908-2936 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 2938-3017, the complete author proof (78 body lines).

Editorial note. Navigation stubs: 10 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 7 of queue_stacks-stacks-more-morphisms.json (pb=4020, stmt 2908-2936, proof 2938-3017).

Editorial note. stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned.

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-refined-valuative-criterion-proper' also exists in stacks-limits, stacks-spaces-limits — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0CMD: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3017–3017. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0ARH: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3017–3017. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CLA: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3017–3017. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:stacks-more-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0CLE: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3017–3017. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CLD: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3017–3017. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CLZ: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3017–3017. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06R7: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3017–3017. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CQK: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3017–3017. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04XH: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3017–3017. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma theorem-chow-finite-type: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2459–2463. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ and $h : \mathcal{U} \to \mathcal{X}$ be morphisms of algebraic stacks. Assume that $\mathcal{Y}$ is locally Noetherian, that f and h are of finite type, that f is separated, and that the image of $|h| : |\mathcal{U}| \to |\mathcal{X}|$ is dense in $|\mathcal{X}|$. If given any 2-commutative diagram*

$$\begin{array}{ccccc} \mathrm{Spec}(K) & \xrightarrow{u} & \mathcal{U} & \xrightarrow{h} & \mathcal{X} \\ {\scriptstyle j}\downarrow & & & & \downarrow{\scriptstyle f} \\ \mathrm{Spec}(A) & & \xrightarrow{y} & & \mathcal{Y} \end{array}$$

where A is a discrete valuation ring with field of fractions K and $\gamma : y \circ j \to f \circ h \circ u$ there exist an extension K'/K of fields, a valuation ring $A' \subset K'$ dominating A such that the category of dotted arrows for the induced diagram

$$\begin{array}{ccc} \mathrm{Spec}(K') & \xrightarrow{} & \mathcal{X} \\ {\scriptstyle j'}\downarrow & \overset{x'}{\nearrow} & \downarrow{\scriptstyle f} \\ \mathrm{Spec}(A') & \xrightarrow{y'} & \mathcal{Y} \end{array}$$

with induced 2-arrow $\gamma' : y' \circ j' \to f \circ x'$ is nonempty (Morphisms of Stacks, Definition 0CLA), then f is proper.

Proof. It suffices to prove that f is universally closed. Let $V \to \mathcal{Y}$ be a smooth morphism where V is an affine scheme. By Properties of Stacks, Lemma 04XH the image I of $|\mathcal{U} \times_{\mathcal{Y}} V| \to |\mathcal{X} \times_{\mathcal{Y}} V|$ is the inverse image of the image of $|h|$. Since $|\mathcal{X} \times_{\mathcal{Y}} V| \to |\mathcal{X}|$ is open (Morphisms of Stacks, Lemma 06R7) we conclude that I is dense in $|\mathcal{X} \times_{\mathcal{Y}} V|$. Also since the category of dotted arrows behaves well with respect to base change (Morphisms of Stacks, Lemma 0CLE) the assumption on existence of dotted arrows (after extension) is inherited by the morphisms $\mathcal{U} \times_{\mathcal{Y}} V \to \mathcal{X} \times_{\mathcal{Y}} V \to V$. Therefore the assumptions of the lemma are satisfied for the morphisms $\mathcal{U} \times_{\mathcal{Y}} V \to \mathcal{X} \times_{\mathcal{Y}} V \to V$. Hence we may assume $\mathcal{Y}$ is an affine scheme.

Assume $\mathcal{Y} = Y$ is an affine scheme. (From now on we no longer have to worry about the 2-arrows γ and γ', see Morphisms of Stacks, Lemma 0CLD.) Then $\mathcal{U}$ is quasi-compact. Choose an affine scheme U and a surjective smooth morphism $U \to \mathcal{U}$. Then we may and do replace $\mathcal{U}$ by U. Thus we may assume that $\mathcal{U}$ is an affine scheme.

Assume $\mathcal{Y} = Y$ and $\mathcal{U} = U$ are affine schemes. By Chow's lemma (Theorem theorem-chow-finite-type) we can choose a surjective proper morphism $X \to \mathcal{X}$ where X is an algebraic space. We will use below that $X \to Y$ is separated as a composition of separated morphisms. Consider the algebraic space $W = X \times_{\mathcal{X}} U$. The projection morphism $W \to X$ is of finite type. We may replace X by the scheme theoretic image of $W \to X$ and hence we may assume that the image of $|W|$ in $|X|$ is dense in $|X|$ (here we use that the image of $|h|$ is dense in $|\mathcal{X}|$, so after this replacement, the morphism $X \to \mathcal{X}$ is still surjective). We claim that for every solid commutative diagram

$$\begin{array}{ccccc} \mathrm{Spec}(K) & \longrightarrow & W & \longrightarrow & X \\ \downarrow & & & \nearrow & \downarrow \\ \mathrm{Spec}(A) & & \longrightarrow & & Y \end{array}$$

where A is a discrete valuation ring with field of fractions K, there exists a dotted arrow making the diagram commute. First, it is enough to prove there exists a dotted arrow after replacing K by an extension and A by a valuation ring in this extension dominating A, see Morphisms of Spaces, Lemma 0ARH. By the assumption of the lemma we get an extension K'/K and a valuation ring $A' \subset K'$ dominating A and an arrow $\mathrm{Spec}(A') \to \mathcal{X}$ lifting the composition $\mathrm{Spec}(A') \to \mathrm{Spec}(A) \to Y$ and compatible with the composition $\mathrm{Spec}(K') \to \mathrm{Spec}(K) \to W \to X$. Because $X \to \mathcal{X}$ is proper, we can use the valuative criterion of properness (Morphisms of Stacks, Lemma 0CLZ) to find an extension K''/K' and a valuation ring $A'' \subset K''$ dominating A' and a morphism $\mathrm{Spec}(A'') \to X$ lifting the composition $\mathrm{Spec}(A'') \to \mathrm{Spec}(A') \to \mathcal{X}$ and compatible with the composition $\mathrm{Spec}(K'') \to \mathrm{Spec}(K') \to \mathrm{Spec}(K) \to X$. Then K''/K and $A'' \subset K''$ and the morphism $\mathrm{Spec}(A'') \to X$ is a solution to the problem posed above and the claim holds.

The claim implies the morphism $X \to Y$ is proper by the case of the lemma for algebraic spaces (Limits of Spaces, Lemma 0CMD). By Morphisms of Stacks, Lemma 0CQK we conclude that $\mathcal{X} \to Y$ is proper as desired. □